

Cebu Upskilling

Upskilling & Career Pathway Application

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ABSTRACT

Unemployment in the Philippines continues to rise, with Central Visayas recording a regional unemployment rate of 3.8 percent in 2025, up from 3.1 percent in 2024, representing approximately 115,000 jobless individuals across the region according to Codis (2026) as reported by Sunstar. Cebu City, despite being the economic center of the Visayas, posted the lowest labor force participation rate in the region at 60.1 percent, indicating that a significant portion of its working-age population is neither employed nor actively seeking work. Nationally, youth unemployment reached 11.7 percent in late 2025, with over 616,000 workers displaced by automation and artificial intelligence-driven restructuring within the same year.

Cebu Upskilling is our solution to that problem. It is an app that asks you what job you want, looks at what skills you already have, and tells you exactly what you are missing and what to study to get there.

The app tracks your progress as you complete courses, updates your profile as your skills grow, and shows you how close you are getting to being ready for your target job. It also connects course providers and employers to the same system, so that the training being offered actually matches what companies are hiring for. The goal is simple: to help more Cebuanos become job-ready, faster, and with a clear plan in hand.

PROBLEM STATEMENT

Unemployment in the Philippines, particularly in Cebu City and the broader Central Visayas region, has been rising at a concerning rate. Job seekers, especially fresh graduates and youth, find themselves unable to compete in a labor market that increasingly demands experience. Despite having educational qualifications, many applicants are rejected due to skill mismatches, lack of hands-on experience, and limited exposure to emerging industries.

"The problem goes beyond unemployment numbers. Even when job opportunities exist, there is a persistent mismatch between available jobs and graduates' skills," CHED Chairperson Dr. Shirley C. Agrupis said, this shows one of the main reasons why unemployment keep rising despite available jobs are on the rise.

Because of this, the problem of unemployment is worsened not by a lack of jobs, but by a lack of job-ready workers. The gap between what employers need and what workers can offer continues to widen. The following basic questions remain unanswered for most unemployed individuals:

1. What specific skills am I missing for my target job?
2. Which training programs or courses are available and appropriate for my current level?
3. How do I track my progress as I upskill?
4. What career paths are realistically accessible based on my background and location?

Without a structured tool to answer these questions, jobseekers rely on guesswork and informal advice rather than data-driven guidance which improves information gathering, resulting in prolonged unemployment, underemployment, and wasted human potential.

SPECIFIC ISSUES IN EXISTING UPSKILLING APPROACHES

Corporate Centered

Corporate has expanded hiring notification and priority making SMEs (Small and Medium Enterprise) being overshadowed, where side hustlers and job seekers who has limited credentials can hardly look for any jobs.

Skills Are Not Properly Connected to Jobs

Right now, skills, courses, and job openings are treated as separate things. A person may finish a course, but there is no clear system showing:

- what jobs that skill leads to
- what salary range to expect
- what next skills to learn

Scattered and Disconnected Resources

Training programs like TESDA courses, DOLE livelihood training, PESO referrals, private review centers, and online learning platforms exist, but they are not connected in one system. Job seekers often need to search multiple websites or offices just to find the right training.

No Personalized Guidance for Learners

Most job placement offices (like PESO) provide general training suggestions, but they don't deeply analyze each person's skill gaps. Because of this, many people are just placed into "common" training programs that may not match what they actually need.

No Long-Term Skill Record

Without a structured record of a user's skill development journey, it is impossible to determine whether an individual is progressing toward employability, stagnating, or regressing. Employers and training providers have no way to verify skill growth over time.

SYSTEM GOAL

- Help users build a clear **skill profile** by taking assessments that show their current skills, experience level, and preferred career path.
- Track each user's **learning progress over time**, including completed courses, certifications, and skill improvements, so they can see how far they've grown.
- Provide **career exploration tools** where users can view job roles, required skills, and salary ranges in Cebu and Central Visayas to help them make informed decisions.
- Not only show high end job availability but also side hustles that greatly benefit small businesses and freelancers.
- Both show the company's job information for application and skill course to satisfy requirements.
- Allow administrators and program managers to generate **reports on skills and job trends**, helping improve training programs and support local workforce planning.
- Match users with the right **learning paths** by comparing their current skills with job requirements and suggesting courses, training, and certifications to close the gaps.

PROPOSED SOLUTION OVERVIEW

The proposed solution is Cebu Upskilling, an application that is designed specifically for upskilling, career guidance, and employment support for both job seekers and the enterprise.

The product also replaces overreliance of guesswork and informal records of current skill attainment to data-driven and systematic output of current skill record, that helps greatly in sorting out field of expertise when job hunting.

This also help enterprises (both large corporate and Small and Medium Enterprises) to faster acquisition of personnels with recommendation coming from the raw information of the applicant's skill set.

How the System Addresses Each Problem

Corporation Centered

- Small Corporation will be able to place their hirings through the app and will be filtered, to avoid being overshadowed with high corporates.
- Job Seekers will be able to filter salary and schedule and choose a job that is compatible with their skills or capability (such as, student who want to side hustle for extra income to satisfy their education needs)

Skills Are Not Properly Connected to Jobs

- Job seekers will be able to see available jobs from job section and scan the required skills. The required skills will be given in the learn area to help the job seekers acquire the necessary skills.
- Enterprises are required to fill what skills and schedules are needed as part of requirement, to avoid absence of information dissemination

Scattered and Disconnected Resources

- Training and courses in the app are heavily based on government agencies, corporate partnerships, and open yet approved sources.
- All of the mentioned sources will be gathered in the app as form of learning courses.

No Personalized Guidance for Learners

- Learners with vast skills can be noted through the app, this serves as quick view of profile and skill set, and can gain more courses that is tailored to the job seeker's current skills (eg. Job seeker has more skill set, intermediate lessons will be given)

No Long-Term Skill Record

- Since the application is online, all data coming from it will be saved through databases (online record) making it easier to save and organize.

MEASUREMENT METHOD

Measurement with no description is harder to evaluate and compute. Henceforth, the system follows scaling as a conversion from description into numerical form, in the same manner as a Likert scale for subjectivity. This can greatly help transform subjective data into numerical forms to help give graphs for visual output.

One of examples that uses a likert scale is:

Level	Label	Description
5	Expert	Full professional mastery; capable of teaching others and leading in the field
4	Advanced	Strong working knowledge; can handle complex tasks independently
3	Intermediate	Functional understanding; can perform standard tasks with minimal guidance
2	Beginner	Basic awareness; requires significant guidance and supervision
1	No Knowledge	No prior exposure or experience with the skill

The experience scale helps the application to classify which courses the learner be able to acquire, they may go to the no knowledge up to technical courses.

The app also conduct examination to the learners to check their progress and verify their learnings. The examination is also supervised with proctor with visual and audio open, and a technology protocol that check current user's screen and opened tabs to avoid cheating.

By judging the score whether the learner can now actively seek job opportunities, the app uses

Job Match Score:

Score Range	Match Level	Description
85–100%	Highly Qualified	User meets or exceeds requirements for the target role
65–84%	Qualified	Minor skill gaps; ready for application with targeted preparation
45–64%	Developing	Moderate gaps; a structured upskilling plan is recommended
25–44%	Early Stage	Significant gaps; foundational courses required before application
0–24%	Not Yet Ready	Major misalignment; long-term pathway required

This scale determines a user's current position relative to their target job's requirements, enabling the system to calculate gap severity and prioritize course recommendations accordingly.

This scale drives the dashboard display for each user, providing a clear, motivating view of their progression toward any selected target career.

USER AND STAKEHOLDERS

System Users

User Type	Role	Decisions Enabled
Learners / Job Seekers	Individuals seeking employment, career change, or skill advancement. Primary end-users of the upskilling application.	Learners use the system to identify specific skill gaps, select appropriate training, track their progress, and gauge their readiness for target job roles.
Course Providers	TESDA accredited centers, partnered corporates, private training institutions, and online learning platforms that supply upskilling content.	Providers can identify which skills have the highest unmet demand in the learner base and adjust their course offerings accordingly.
Employers	Companies and hiring organizations that post role requirements and use the system to identify pipeline candidates.	Employers can forecast talent availability by skill domain, identify gaps in the regional workforce pipeline, and inform upskilling providers of priority skill demands.
Administrators	System administrators who maintain reference data, manage accounts, and ensure data integrity.	Administrators ensure skill taxonomies are current and accurately mapped to labor market demands, user accounts are correctly assigned, and data quality issues are resolved promptly.

External Stakeholders

- DOLE and TESDA: Can use the system's reports to identify which skills are most needed in the workforce, helping them improve training programs and make better funding decisions.
- Local Government Units (LGUs) and Barangay Offices: Can view employment and skills data in their communities, allowing them to create better livelihood and job assistance programs.
- Schools, colleges, and researchers: Can study anonymous workforce trends and skill gaps to better understand employment challenges and recommend solutions.
- Employers: Benefit from a larger pool of job seekers who have relevant skills and certifications that match industry needs.
- Job seekers and underemployed individuals: Gain access to personalized learning paths, skill development opportunities, and better chances of finding suitable employment.

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